Assignment:

Exploring Risk Management

Duration: 60 Minutes

Learning Objectives

Risk identification is an iterative activity that happens throughout the project. By the end of this lab you'll be able to

- Identify pure risks and business risks
- Work with the project team and stakeholders to identify risk events
- Create a risk register

Identify Project Risks

You are the project manager of the HGN Project for your organization. This project will create a log cabin in the Blue Ridge Mountains of Georgia. According to the project scope and requirements, the site where the cabin will be created will require a water well for the cabin's water supply and solar equipment and windmills for electricity. The cabin is a two-bedroom home of 2,000 square feet isolated on a ten-acre plot of land. The architects, electrical engineers, and designers have created the plans for a self-sustaining mountain cabin for your client, now it's up to you and your project team to begin risk identification.

1. What documents would you like to review in order to identify risks within the project?

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2. Are there are any pure risks that you can identify given this information?

3. What business risks are revealed in this scenario?

4. Your risk identification process is taking place with the project team and the architects, electrical engineers, and designers. Why should these people be involved with the risk identification process?

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5. Mike, one of your project team members, has identified a risk dealing with the windmills the property will use for electricity. Mike points out that no one on the project team has actually ever constructed a windmill before, although the vendor that supplies the windmill provides detailed directions for assembling the device. The electrical engineers tell Mike that the windmill is relatively easy to install; the process isn't much different than installing the solar panels as Mike has done in the past. Does the windmill construction and installation present any risk in this project? Why or why not?

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6. As the project manager of this project, can you create five questions for the people in this risk identification meeting to help you identify risks in this project?

7. As you and the participants in the risk identification meeting discover risks about the project, Sam is recording the risk events for the risk register. What is the risk register, and what information would you want for each event in the risk register?

Complete Qualitative Risk Analysis

You are serving as the project manager of the GHY Project for your company. This project will create a new computer network in a historical warehouse that's been converted to an office space. The project will require 160 network connections to be installed throughout the new office building, 120 new workstations to be installed and configured for local area network and Internet connectivity. In addition there will be twenty network-based printers and five file servers users will need to connect to. You've been working with your project team and the vendors to identify the risks in the project, and your team has identified several risks. Here are ten risks in your project to review:

- The vendor may be late with materials.
- The maximum distance of network cable from central patch panel may be excessive.
- The brick building structure may make it difficult to install wiring.
- There is no drop ceiling (plenum) for cables.
- The ability to meet deadlines is in doubt, considering the challenges of the project.
- The office building has historical restrictions on construction.
- Physical security of the equipment during the project may prove inadequate.
- The building owner must inspect the work, which could cause additional delays or rework.
- The network printers need to be centrally located, but with network cables out of sight.
- A secured server room will need to be constructed first, but the room will need approval for power, air conditioning, and the physical structure by the building owner.

1. All of these risks need to pass through qualitative analysis. You can, for this lab, search the Internet, interview a subject matter expert, or base your analysis on your experience. In the following table you'll determine the probability of the risk event, assess the impact of the risk event, and create a risk score on an ordinal scale of very low to very high. As an example, the first risk event has been completed for you:

Identified Risk	Probability	Impact	Risk Score
Vendor lateness with materials	Moderate	Very high	High
Maximum distance of network cable from central patch panel			
Brick building structure may prove difficult to install wiring			
There is no drop ceiling (plenum) for cables			
Ability to meet deadline considering challenges of project			
Office building has historical restrictions on construction			
Physical security of the equipment during the project			
Building owner must inspect the work, which could cause additional delays or rework			
The network printers need to be centrally located, but with network cables out of sight			
A secured server room will need to be constructed first, but the room will need approval for power, air conditioning, and the physical structure by the building owner			
Vendor lateness with materials	Moderate	Very high	High
Maximum distance of network cable from central patch panel			

2. Based on your qualitative analysis, what risk events should move on to quantitative analysis?

3. In your analysis, which risk events are most threatening to the project?

4. What is the danger of using only qualitative analysis?

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Learning Objectives

Quantitative risk analysis comes after qualitative risk analysis and quantifies the project's risk exposure. By the end of this lab you'll be able to

- Create a quantitative probability-impact matrix
- Determine the project's risk exposure
- Recommend a contingency reserve for the project

Complete Quantitative Risk Analysis

You are serving as the project manager of the GHY Project for your company. This project will create a new computer network in a historical warehouse that's been converted to an office space. The project will require 160 network connections to be installed throughout the new office building and 120 new workstations to be installed and configured for local area network and Internet connectivity. In addition, there will be twenty network-based printers and five file servers users will need to connect to.

All of the project risks have passed through qualitative analysis, and these now need to, for this exercise, pass through quantitative analysis. Your project team has interviewed subject matter experts, has completed testing of the risk events, and are now ready to rank the results of their tests. Complete the following table to find the expected monetary value for each risk event.

Identified Risk	Probability	Impact	Expected Monetary Value
Vendor lateness with materials	0.50	–$15,000	
Maximum distance of network cable from central patch panel	0.25	–9,000	
Brick building structure may prove difficult to install wiring	0.80	–45,000	
There is no drop ceiling (plenum) for cables	0.70	–35,000	
Ability to meet deadline considering challenges of project	0.40	–12,000	
Office building has historical restrictions on construction	0.65	–12,000	
Physical security of the equipment during the project	0.20	–75,000	
Building owner must inspect the work which could cause additional delays or rework	0.60	–5,000	
The network printers need to be centrally located, but with network cables out of sight	0.55	–14,000	
A secured server room will need to be constructed first, but the room will need approval for power, air conditioning, and the physical structure by the building owner	0.90	–45,000	

1. What are three risk events with the largest expected monetary value?

2. What is the project’s risk exposure?

3. What should you request as this project's contingency reserve?

4. For planning, rank the risk events from high to low:

Create Risk Responses

1. Mary is a project manager of a construction project. There are many dangerous activities in the project, and she wants to be certain her construction crew is safe as they complete the project work. In this scenario what’s an example of risk avoidance for the pure risks in the project?

2. Search the Internet and find a business news story of two companies using the share risk response to seize an opportunity. Document the story and web site here:

3. John is the project manager of a jazz festival in your hometown. This jazz festival is a major fundraiser for the arts museum. Because the jazz festival is held in the springtime, John recommends that the project purchase weather insurance. Weather insurance is an example of what type of risk response?

4. What is an example of positive risk that could use the exploit risk response?

5. Stephanie is the project manager of an IT project to create a web-based catalog for a bicycle manufacturer. Stephanie's project will document all bikes the organization creates, including videos, photos, and user stories about the bikes. The project will also coordinate orders from resellers of the bikes and the manufacturing facility. Stephanie stresses to the project team that they need to address the risk of lost orders, data loss, and faulty equipment. If the team wanted to mitigate this risk, what response would you recommend?

6. Ann is the project manager of a manufacturing project to create plastic sunglasses for a client. The project customer has promised a bonus if they can deliver the sunglasses at least ten days early and with zero defects. Ann knows this can be done if the project team will work just one hour extra per day for the next week. The project team agrees to do the overtime work in exchange for a share of the bonus. How is this scenario an example of enhancing?

Solution: Identifying Project Risks

1. What documents would you like to review in order to identify risks within the project?

   Risk identification requires a thorough review of the project documents to help uncover risks in the project. You'll rely on eleven inputs for this process: risk management plan, activity cost estimates, activity duration estimates, scope baseline, stakeholder register, cost management plan, schedule management plan, quality management, enterprise environmental factors, organizational process assets, and project documentation. The project documentation can include the assumptions log, performance reports, the results of earned value, and other supporting detail for the project planning.

2. Are there are any pure risks that you can identify given this information?

   Pure risks are risk events that only have a downside. In construction there are always safety risks that must be addressed. There may also be risks with the electrical work, fire, and theft, and the remoteness of the work could create hazardous situations for the project team.

3. What business risks are revealed in this scenario?

   A business risk is a risk event that could have a positive side or a downside. Business risks could include completing or not completing the project, making a profit on the project, learning new things about the project materials, and constructing the project in the remote areas that may lift some regulatory guidelines the project team may have to work with otherwise.

4. Your risk identification process is taking place with the project team and the architects, electrical engineers, and designers. Why should these people be involved with the risk identification process?

   The project manager definitely wants the project team, the architects, the electrical engineers, and the designers to help identify the risk events, as these people are closest to the project work. They understand the pure and business risks associated with the project, and they'll be involved with the project execution. By including these people in this process, you're helping them find ownership of the risk in the project.

5. Mike, one of your project team members, has identified a risk dealing with the windmills the property will use for electricity. Mike points out that no one on the project team has actually ever constructed a windmill before, although the vendor that supplies the windmill provides detailed directions for assembling the device. The electrical engineers tell Mike that the windmill is relatively easy to install; the process isn't much different than installing the solar panels as Mike has done in the past. Does the windmill construction and installation present any risk in this project? Why or why not?

   Yes, this is a risk in the project. While Mike may have reports of the ease of the work, a person never knows how challenging a task is until he actual does the work. While the activity details and responses may later be addressed, at this point there's uncertainty surrounding the activity, so this is definitely a risk event for the project.

6. As the project manager of this project, can you create five questions for the people in this risk identification meeting to help you identify risks in this project?

   There are many questions the project manager could ask to help the project team and subject matter experts identify risks. Your answers will likely be different than what's provided here, but these are five sample questions the project manager could ask the participants of the risk identification meeting:

   - What materials and activities have we never tried before that we'll be doing in this project?
   - What's the most dangerous activity in the project?
   - What activities will likely be the most difficult to complete?
   - What activities are reliant on vendors?
   - What deliverables will require the coordination of more than three people on the project team?

7. As you and the participants in the risk identification meeting discover risks about the project, Sam is recording the risk events for the risk register. What is the risk register, and what information would you want for each event in the risk register?

   The risk register is a central repository of all risks in the project. Each identified risk should be entered into the risk register, its characteristics should be documented, and the person who'll be the risk owner should also be documented. As the project moves forward, the risk register should be updated to reflect the characteristics of any changed risk events. Whenever risk events are identified, they should be added to the risk register.

Solution: Completing Qualitative Analysis

1. All of these risks need to pass through qualitative analysis. You can, for this lab, search the Internet, interview a subject matter expert, or base your analysis on your experience. In the following table you'll determine the probability of the risk event, assess the impact of the risk event, and create a risk score on an ordinal scale of very low to very high. Your answers will likely be different than what's been provided here:

Identified Risk	Probability	Impact	Risk Score
Vendor lateness with materials	Moderate	Very high	High
Maximum distance of network cable from central patch panel	Low	Very High	Moderate
Brick building structure may prove difficult to install wiring	High	Very High	High
There is no drop ceiling (plenum) for cables	High	High	High
Ability to meet deadline considering challenges of project	High	High	High
Office building has historical restrictions on construction	Moderate	High	Moderate
Physical security of the equipment during the project	Low	Low	Low
Building owner must inspect the work which could cause additional delays or rework	Moderate	High	Moderate
The network printers need to be centrally located, but with network cables out of sight	High	Moderate	Moderate
A secured server room will need to be constructed first, but the room will need approval for power, air conditioning, and the physical structure by the building owner	High	High	High

2. Based on your qualitative analysis, what risk events should move on to quantitative analysis?

   Your answers are likely very different than what has been provided. Usually, risks that have a moderate score or higher should move on to quantitative analysis.

3. In your analysis, which risk events are most threatening to the project?

   Any risks even with a qualitative analysis score of high or very high are considered threatening to the project.

4. What is the danger of using only qualitative analysis?

   Qualitative analysis is subjective, is fast, and doesn't provide much supporting detail for the scoring. This exercise, albeit a simple look into qualitative analysis, is proof of the subjective nature of this approach to risk analysis. The scores you created are likely quite different than what's been provided here or how other readers of this book may have scored the risk events. Qualitative analysis is based on experience, intuition, and some light research—which lends itself to fast, topical reviews of potentially hundreds of risk events. The danger of this approach is that some risk events may need to be analyzed more seriously than what time or the methodology allows.

Solution: Performing Quantitative Analysis

All of the project risks have passed through qualitative analysis, and these now need to, for this exercise, pass through quantitative analysis. Your project team has interviewed subject matter experts, has completed testing of the risk events, and are now ready to rank the results of their tests. Complete the following table to find the expected monetary value for each risk event.

Identified Risk	Probability	Impact	Expected Monetary Value
Vendor lateness with materials	0.50	–$15,000	–$7,500
Maximum distance of network cable from central patch panel	0.25	–9,000	–2,250
Brick building structure may prove difficult to install wiring	0.80	–45,000	–36,000

Identified Risk	Probability	Impact	Expected Monetary Value
There is no drop ceiling (plenum) for cables	0.70	–35,000	–24,500
Ability to meet deadline considering challenges of project	0.40	–12,000	–4,800
Office building has historical restrictions on construction	0.65	–12,000	–7,800
Physical security of the equipment during the project	0.20	–75,000	–15,000
Building owner must inspect the work which could cause additional delays or rework	0.60	–5,000	–3,000
The network printers need to be centrally located, but with network cables out of sight	0.55	–14,000	–7,700
A secured server room will need to be constructed first, but the room will need approval for power, air conditioning, and the physical structure by the building owner	0.90	–45,000	–40,500

1. What are three risk events with the largest expected monetary value?

    - A secured server room will need to be constructed first, but the room will need approval for power, air conditioning, and the physical structure by the building owner (–$40,500)
    - Brick building structure may prove difficult to install wiring (–$36,000)
    - There is no drop ceiling (plenum) for cables (–$24,500)

2. What is the project's risk exposure?

    The project's risk exposure is the sum of the expected monetary value. In this project the exposure is –$149,050.

3. What should you request as this project’s contingency reserve?

The contingency reserve should offset the risk exposure for the project. In this project the value should be $149,050.

4. For planning, rank the risk events from high to low:

Identified Risk	Probability	Impact	Expected Monetary Value
A secured server room will need to be constructed first, but the room will need approval for power, air conditioning, and the physical structure by the building owner	0.90	–$45,000.00	–$40,500.00
Brick building structure may prove difficult to install wiring	0.80	–45,000.00	–36,000.00
There is no drop ceiling (plenum) for cables	0.70	–35,000.00	–24,500.00
Physical security of the equipment during the project	0.20	–75,000.00	–15,000.00
Office building has historical restrictions on construction	0.65	–12,000.00	–7,800.00
The network printers need to be centrally located, but network with cables out of sight	0.55	–14,000.00	–7,700.00
Vendor lateness with materials	0.50	–15,000.00	–7,500.00
Ability to meet deadline considering challenges of project	0.40	–12,000.00	–4,800.00
Building owner must inspect the work which could cause additional delays or rework	0.60	–5,000.00	–3,000.00
Maximum distance of network cable from central patch panel	0.25	–9,000.00	–2,250.00

Solution: Planning Risk Responses

Your solutions for this lab may be slightly different than what appears here, but these are the ideal solutions:

1. Mary is a project manager of a construction project. There are many dangerous activities in the project, and she wants to be certain her construction crew is safe as they complete the project work. In this scenario what's an example of risk avoidance for the pure risks in the project?

   A pure risk is any risk that has only a downside. In this example, someone getting injured or even loss of life are pure risks. In some projects, such as in construction, it's impossible to avoid the risks and to still complete the project work. Training, safety equipment, even purchasing insurance would be appropriate strategies, but these risk responses don't actually avoid the existing risk events. Not accepting the project work is an example of avoidance, but then Mary could not complete the project scope.

2. Search the Internet and find a business news story of two companies using the share risk response to seize an opportunity. Document the story and web site here:

   There are many possible solutions to this question. Your solution should include two other competing companies that have partnered together to seize an opportunity. Any joint-venture or teaming agreement is an example of the shared risk response.

3. John is the project manager of a jazz festival in your hometown. This jazz festival is a major fundraiser for the arts museum. Because the jazz festival is held in the springtime, John recommends that the project purchase weather insurance. Weather insurance is an example of what type of risk response?

   Weather insurance, and really any insurance, is an example of risk transference. Remember, transference does not eliminate the risk but only transfers ownership of the risk to someone else. In this example there's still the risk that it could rain, but the insurance company now owns the risk, not John's organization or the jazz festival.

4. What is an example of positive risk that could use the exploit risk response?

   A positive risk using the exploit risk response is a scenario where the project manager is trying to take advantage of an opportunity. The project manager in your solution should be trying to make the positive risk, or opportunity, come true.

5. Stephanie is the project manager of an IT project to create a web-based catalog for a bicycle manufacturer. Stephanie's project will document all bikes the organization creates, including videos, photos, and user stories about the bikes. The project will also coordinate orders from resellers of the bikes and the manufacturing facility. Stephanie stresses to the project team that they need to address the risk of lost orders, data loss, and faulty equipment. If the team wanted to mitigate this risk, what response would you recommend?

   Mitigation is a reduction in the probability and/or impact of the risk event. In this scenario Stephanie and her project team could use fault-tolerant computer devices, redundancy of data, and regular data backups to mitigate the risk event. There are likely other solutions that you've created, which are fine as long as your solutions reduced the probability and/or impact of the risk event.

6. Ann is the project manager of a manufacturing project to create plastic sunglasses for a client. The project customer has promised a bonus if they can deliver the sunglasses at least ten days early and with zero defects. Ann knows this can be done if the project team will work just one hour extra per day for the next week. The project team agrees to do the overtime work in exchange for a share of the bonus. How is this scenario an example of enhancing?

   Enhancing tries to increase the positive impacts of an opportunity within the project. By the project team working additional time on a risk event, they are enhancing the likelihood that they'll be able to realize the accelerated deadline. Note that this scenario is similar to the exploiting risk response, but there's a subtle difference. Exploiting takes steps to ensure the positive risk happens, whereas enhancing tries to make the positive risk happen, but it can't ensure that it will happen. In this scenario the project team could work the overtime, but they could still have errors or defects that would cause the project to be late.